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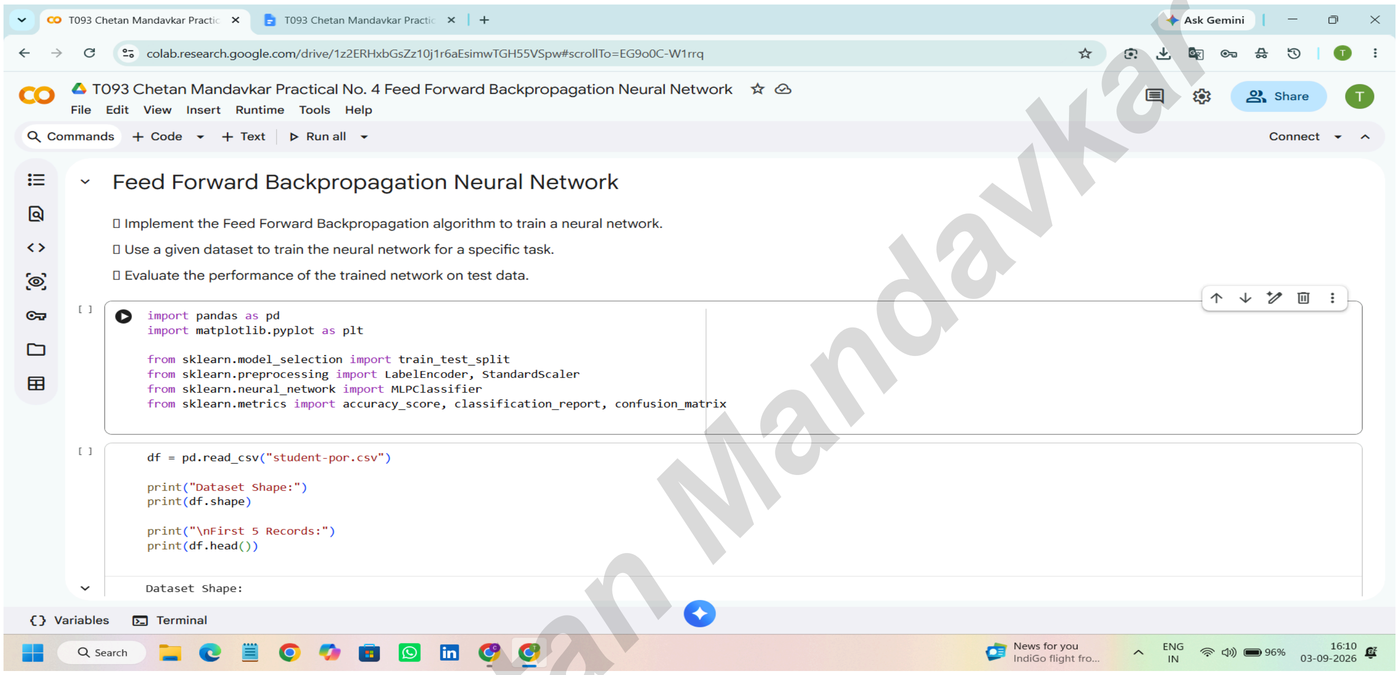
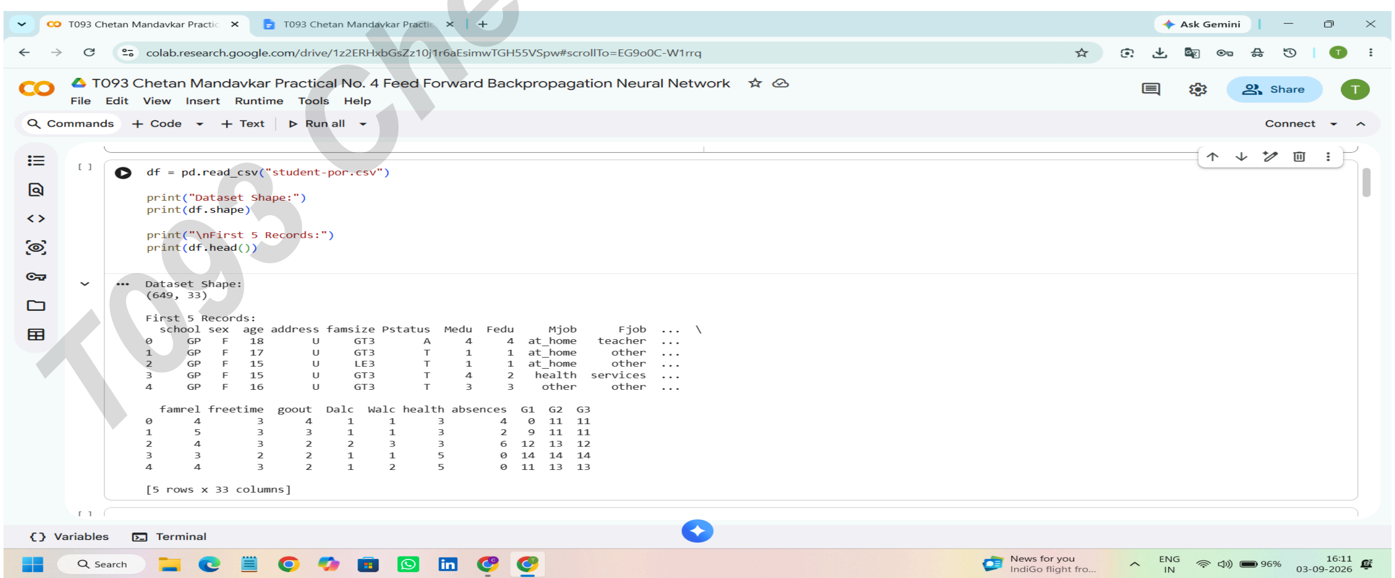
Aim:- Feed Forward Backpropagation Neural Network
Implement the Feed Forward Backpropagation algorithm to train a neural network. Use a given dataset to train the neural network for a specific task. Evaluate the performance of the trained network on test data





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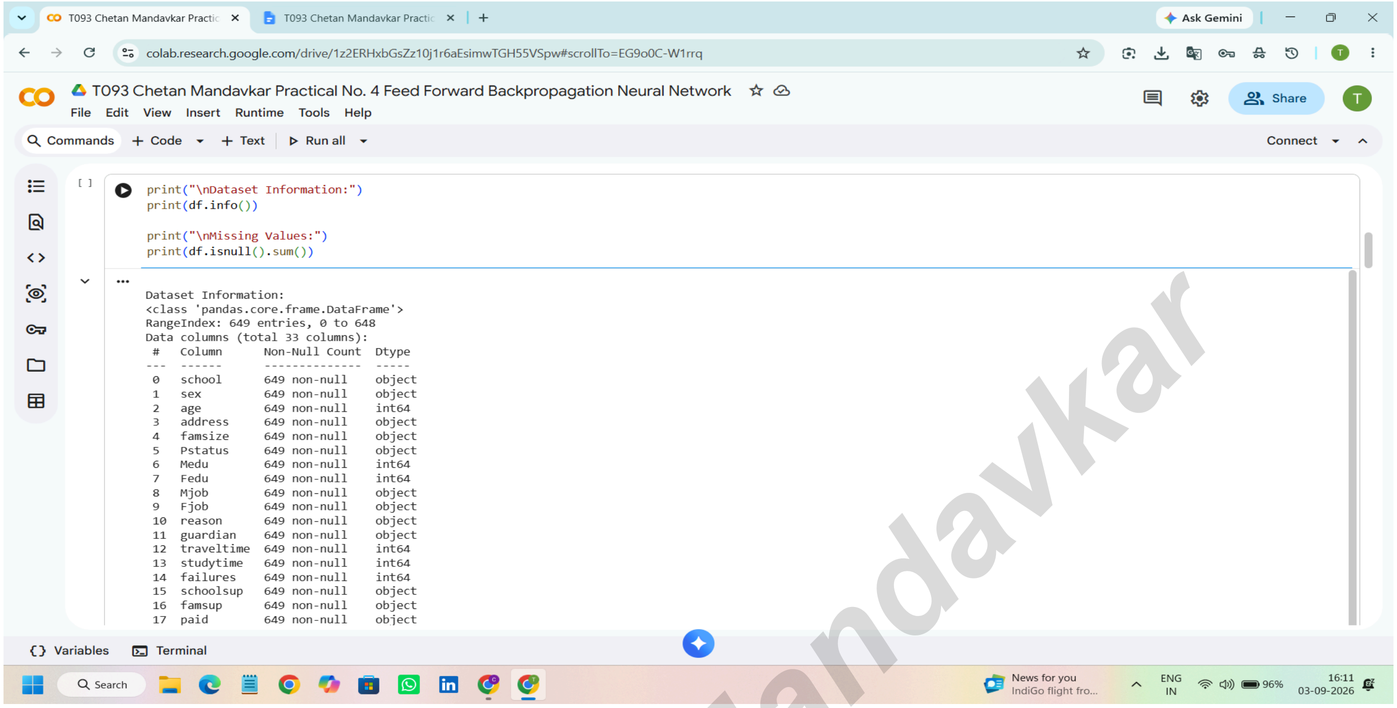
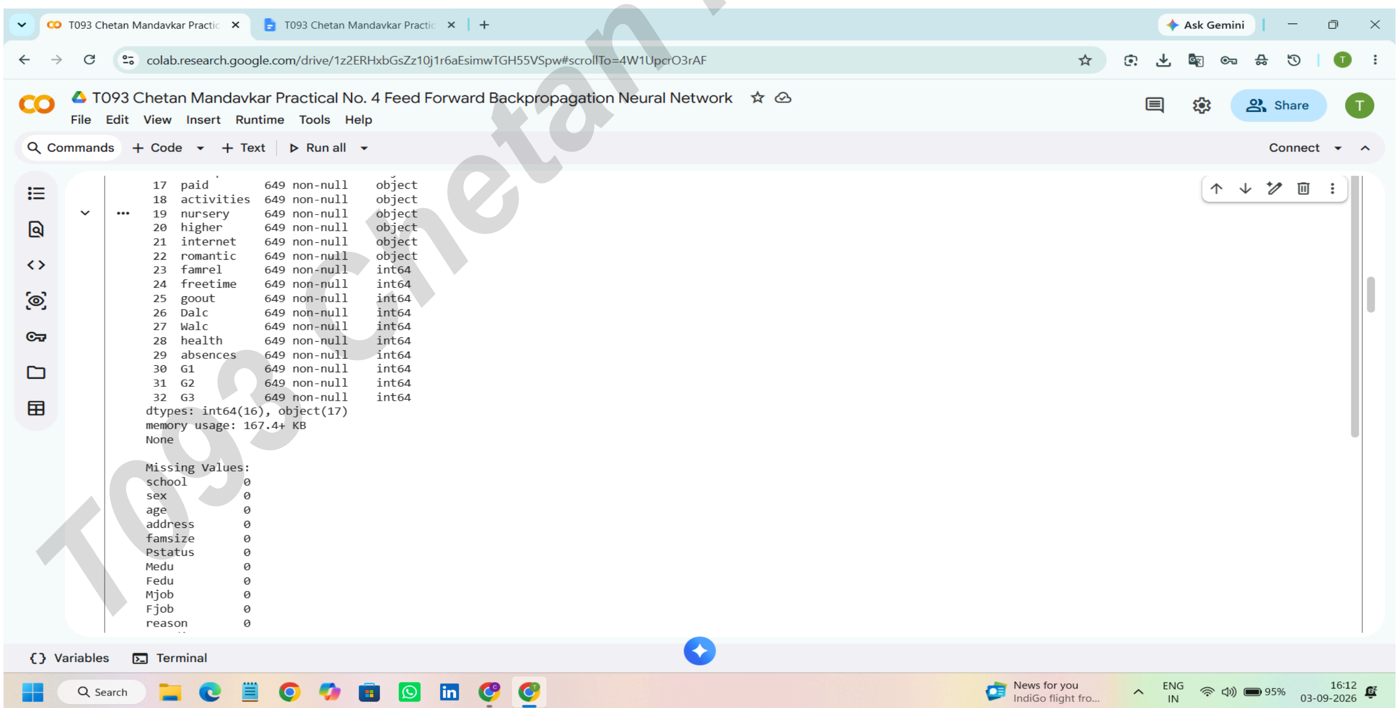



```
print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 649 entries, 0 to 648
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      649 non-null    object
 1   sex         649 non-null    object
 2   age         649 non-null    int64
 3   address     649 non-null    object
 4   famsize     649 non-null    object
 5   Pstatus     649 non-null    object
 6   Medu        649 non-null    int64
 7   Fedu        649 non-null    int64
 8   Mjob        649 non-null    object
 9   Fjob        649 non-null    object
 10  reason      649 non-null    object
 11  guardian    649 non-null    object
 12  traveltime  649 non-null    int64
 13  studytime   649 non-null    int64
 14  failures    649 non-null    int64
 15  schoolsup   649 non-null    object
 16  famsup      649 non-null    object
 17  paid        649 non-null    object
```




```
 17  paid        649 non-null    object
 18  activities  649 non-null    object
 19  nursery     649 non-null    object
 20  higher      649 non-null    object
 21  internet    649 non-null    object
 22  romantic    649 non-null    object
 23  famrel      649 non-null    int64
 24  freetime    649 non-null    int64
 25  goout       649 non-null    int64
 26  Dalc        649 non-null    int64
 27  Walc        649 non-null    int64
 28  health      649 non-null    int64
 29  absences    649 non-null    int64
 30  G1          649 non-null    int64
 31  G2          649 non-null    int64
 32  G3          649 non-null    int64
dtypes: int64(16), object(17)
memory usage: 167.4+ KB
None

Missing Values:
school        0
sex           0
age           0
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
```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```
age
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
guardian      0
traveltime    0
studytime     0
failures      0
schoolsup     0
famsup        0
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64
```

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```
df['Result'] = df['G3'].apply(
    lambda x: 1 if x >= 10 else 0
)

print("\nResult Distribution:")
print(df['Result'].value_counts())
```

```
Result Distribution:
Result
1    549
0    100
Name: count, dtype: int64
```

```
features = [
    'school',
    'sex',
    'age',
    'address',
    'famsize',
    'Pstatus',
    'Medu',
    'Fedu',
    'Mjob',
    'Fjob',
    'reason',
    'guardian',
    'traveltime',
    'studytime',
```


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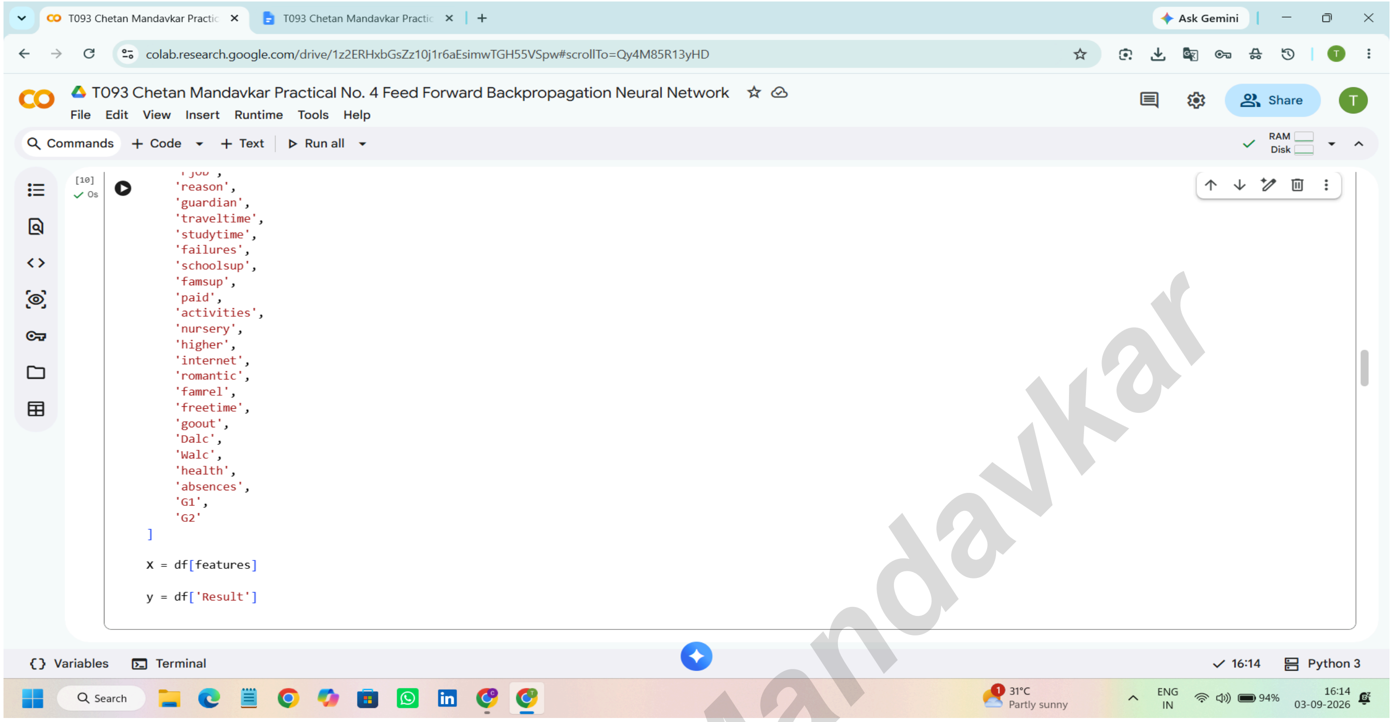
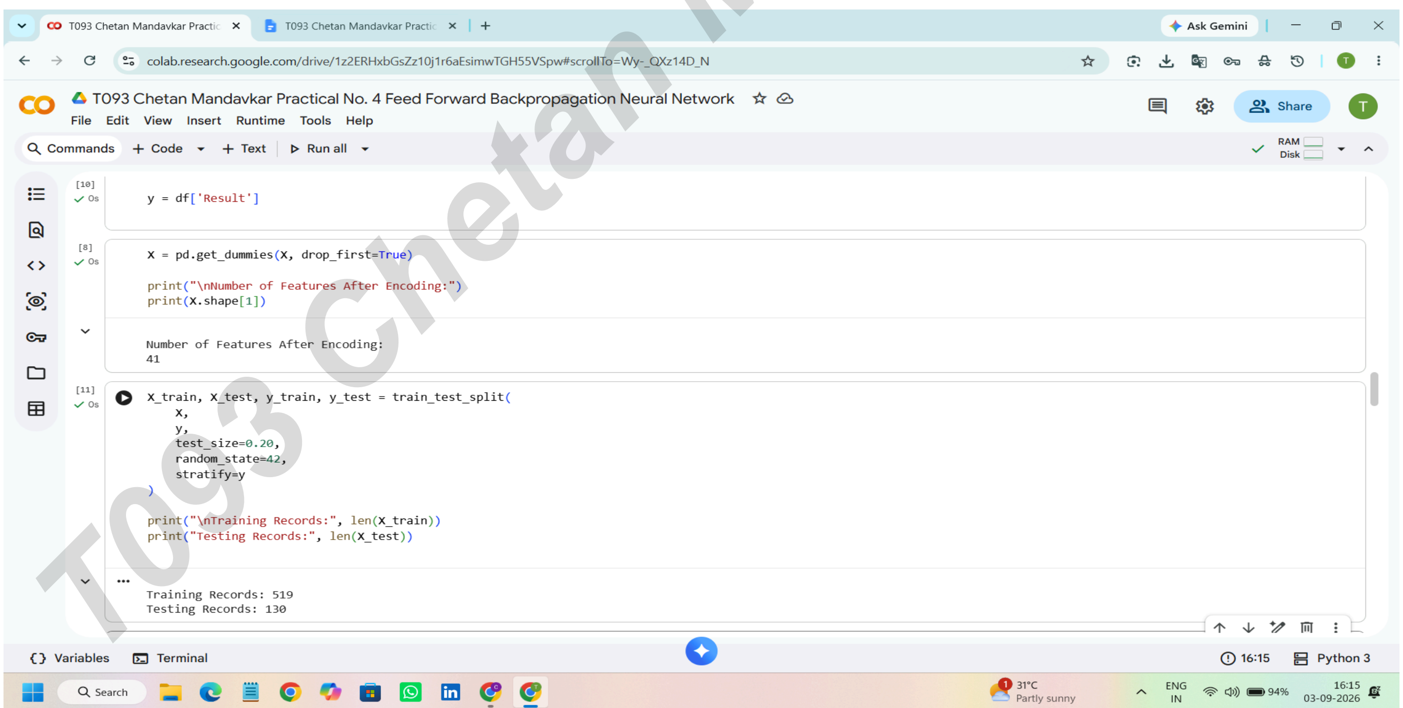


```python
        'job',
        'reason',
        'guardian',
        'traveltime',
        'studytime',
        'failures',
        'schoolsup',
        'famsup',
        'paid',
        'activities',
        'nursery',
        'higher',
        'internet',
        'romantic',
        'famrel',
        'freetime',
        'goout',
        'Dalc',
        'Walc',
        'health',
        'absences',
        'G1',
        'G2'
]

X = df[features]

y = df['Result']
```



```python
y = df['Result']
```

```python
X = pd.get_dummies(X, drop_first=True)

print("\nNumber of Features After Encoding:")
print(X.shape[1])
```

```
Number of Features After Encoding:
41
```

```python
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTraining Records:", len(X_train))
print("Testing Records:", len(X_test))
```

```
Training Records: 519
Testing Records: 130
```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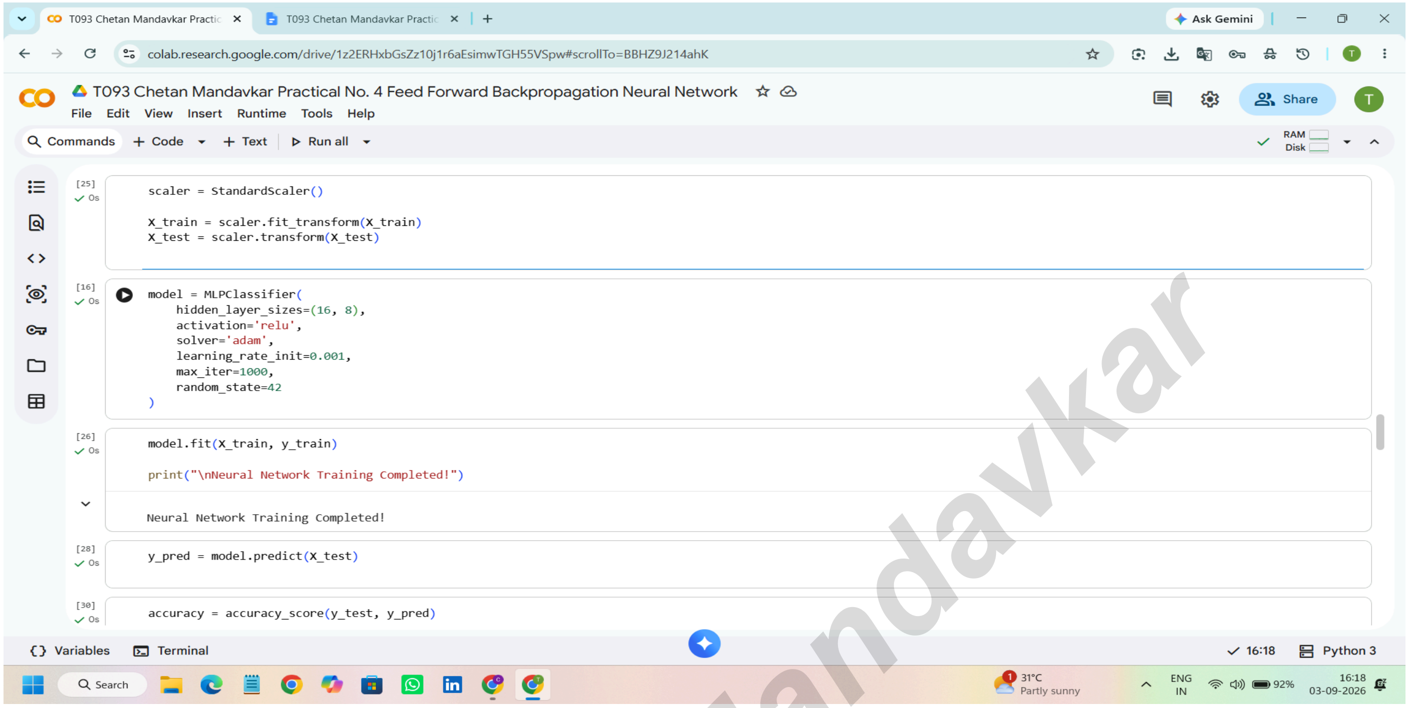


```
scaler = StandardScaler()

x_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = MLPClassifier(
    hidden_layer_sizes=(16, 8),
    activation='relu',
    solver='adam',
    learning_rate_init=0.001,
    max_iter=1000,
    random_state=42
)
```

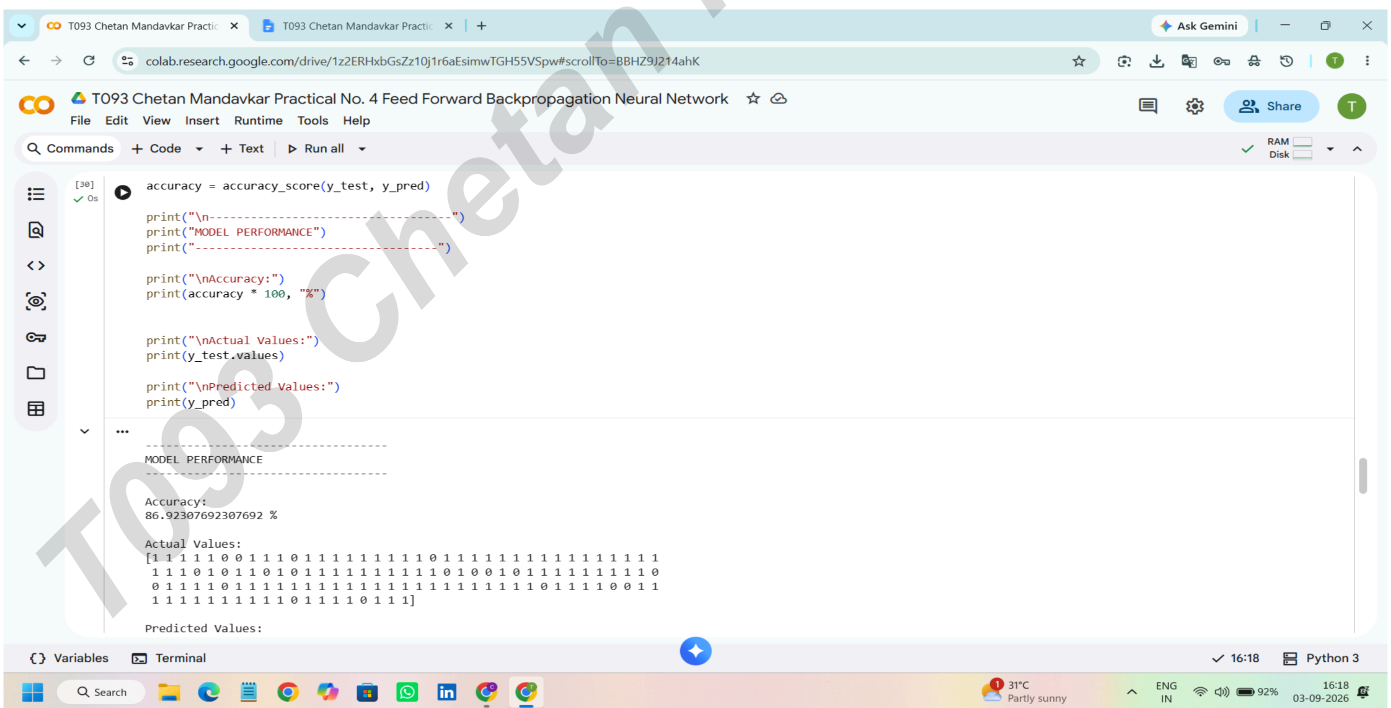
```
model.fit(X_train, y_train)

print("\nNeural Network Training Completed!")
```

```
Neural Network Training Completed!
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```



```
accuracy = accuracy_score(y_test, y_pred)

print("\n-------------------------------")
print("MODEL PERFORMANCE")
print("-------------------------------")

print("\nAccuracy:")
print(accuracy * 100, "%")


print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)
```

```
-----------------------------------
MODEL PERFORMANCE
-----------------------------------

Accuracy:
86.92307692307692 %

Actual Values:
[1 1 1 1 1 0 0 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 0 1 0 1 1 0 1 0 1 1 1 1 1 1 1 1 1 1 0 1 0 0 1 0 1 1 1 1 1 1 1 1 1 0
 0 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 0 0 1 1
 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 0 1 1 1]

Predicted Values:
```

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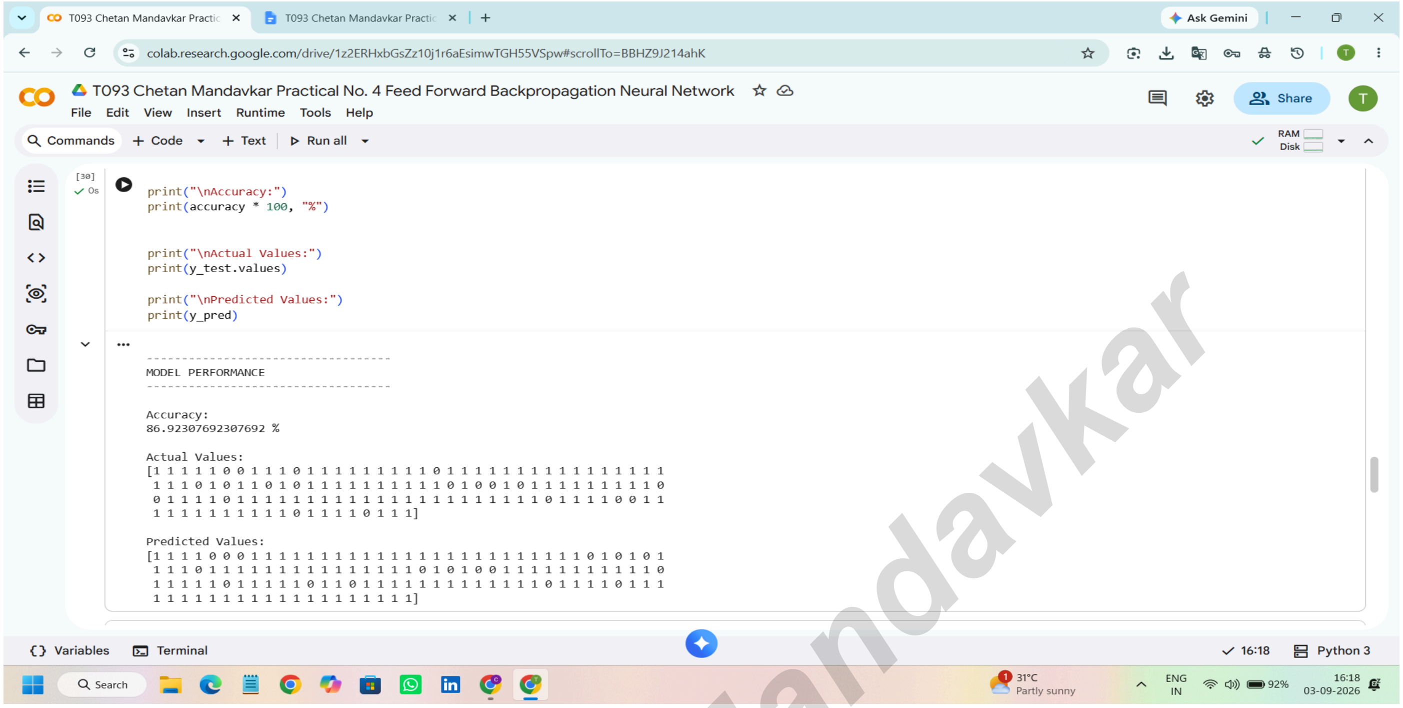

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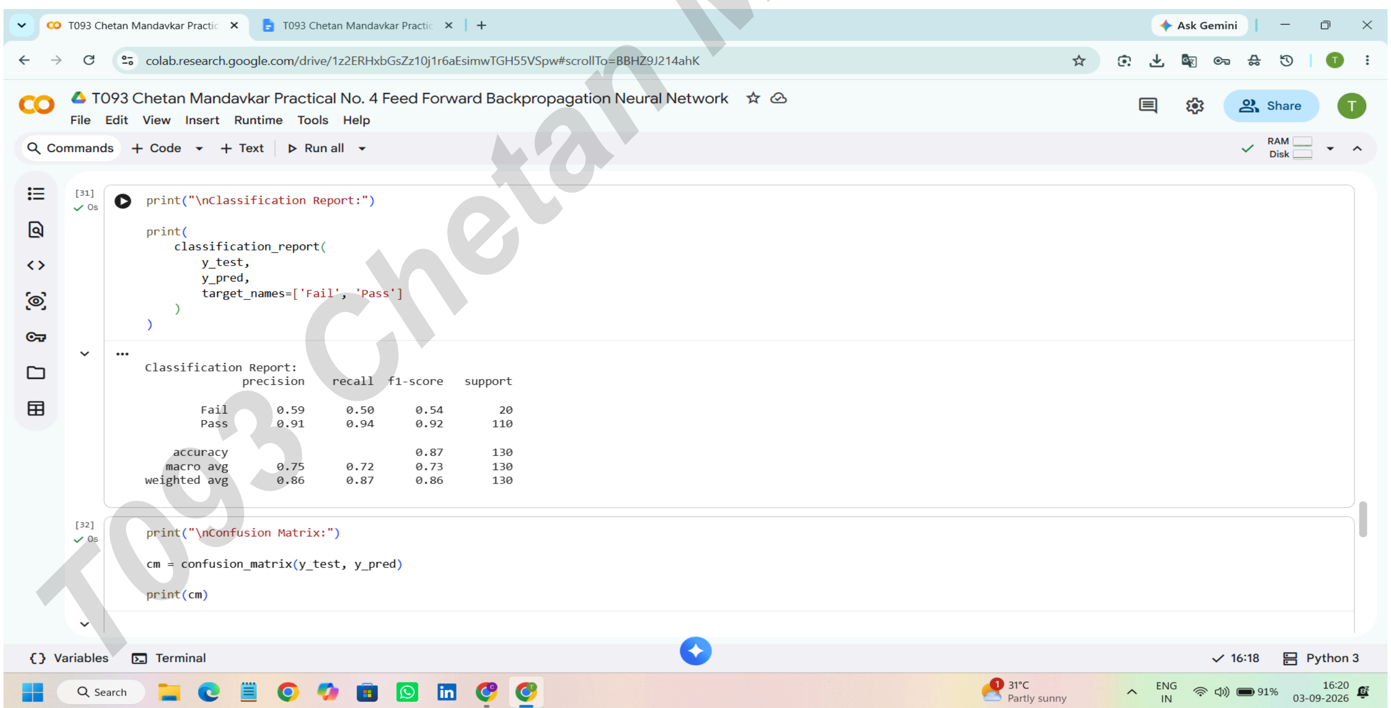
```
print("\nAccuracy:")
print(accuracy * 100, "%")


print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)
```

```
----------------------------------
MODEL PERFORMANCE
----------------------------------

Accuracy:
86.92307692307692 %

Actual Values:
[1 1 1 1 1 0 0 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 0 1 0 1 1 0 1 0 1 1 1 1 1 1 1 1 1 0 1 0 0 1 0 1 1 1 1 1 1 1 1 1 0
 0 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 0 0 1 1
 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 0 1 1 1]

Predicted Values:
[1 1 1 1 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 0 1 0 1
 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 0 1 0 0 1 1 1 1 1 1 1 1 1 1 0
 1 1 1 1 1 0 1 1 1 1 1 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 0 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
```



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```
print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=['Fail', 'Pass']
    )
)
```

```
Classification Report:
              precision    recall  f1-score   support

        Fail       0.59      0.50      0.54        20
        Pass       0.91      0.94      0.92       110

    accuracy                           0.87       130
   macro avg       0.75      0.72      0.73       130
weighted avg       0.86      0.87      0.86       130
```

```
print("\nConfusion Matrix:")

cm = confusion_matrix(y_test, y_pred)

print(cm)
```


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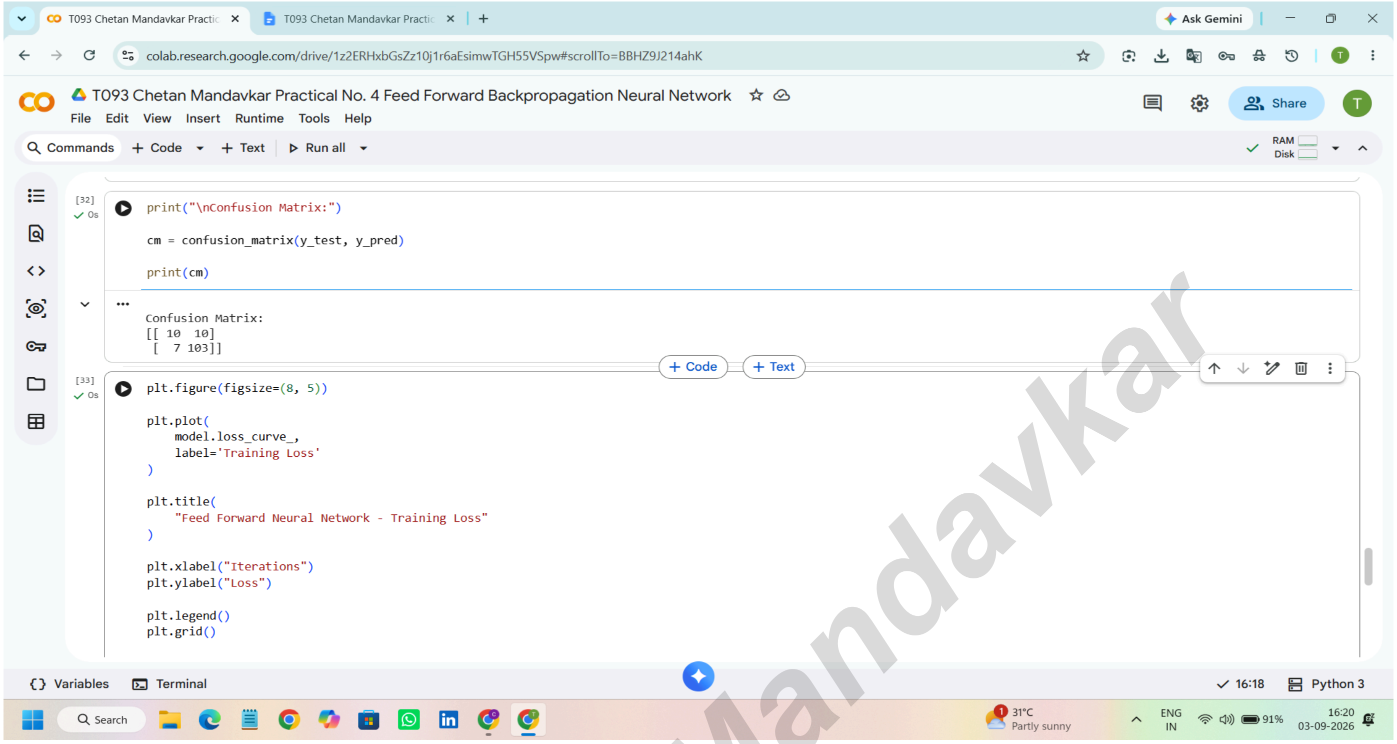

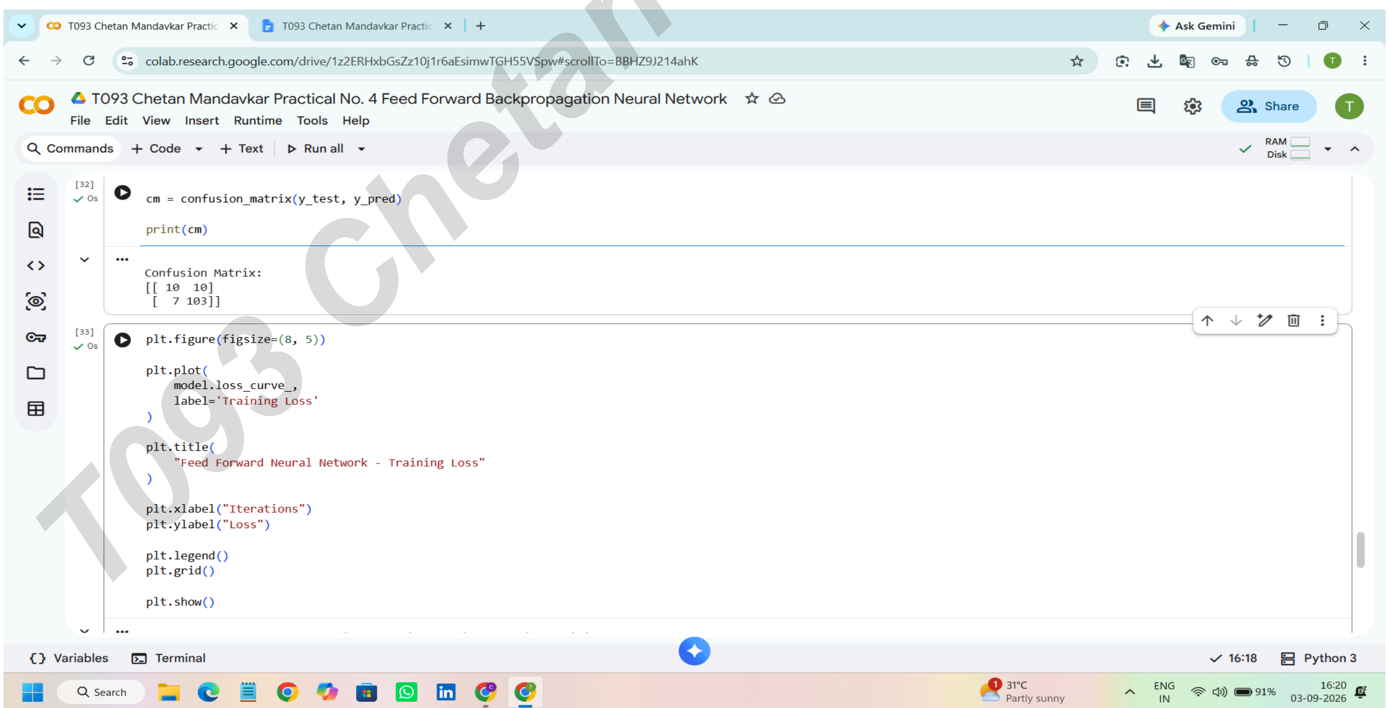

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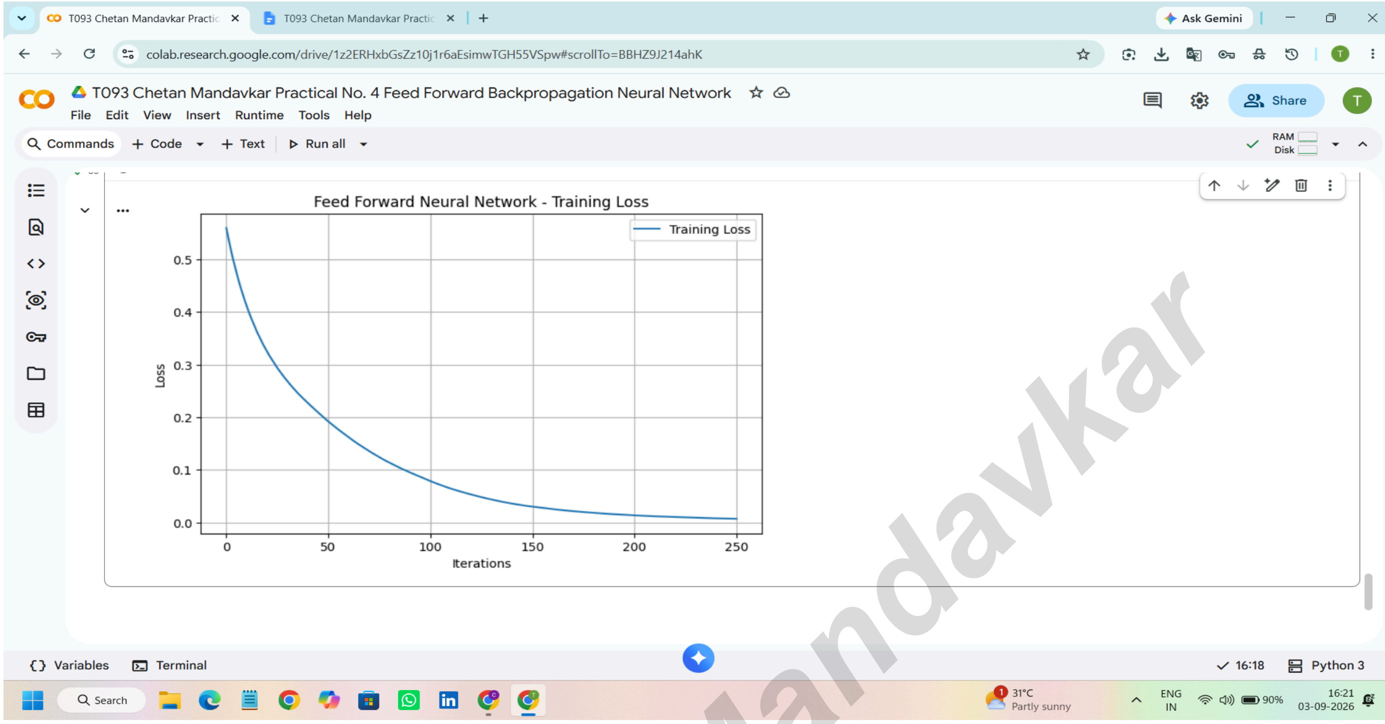

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